HYDROSTAR SYSTEMS INTEGRATION REPORT

Project: Yellow Creek Ravine E.coli Active Modeling & Coherent Restoration
Technical Specifications: 10-Year CFU Projections & Exact QK-GPR Model Architecture

Executive Summary

This technical document presents the mathematical and computational framework for predicting and mitigating fecal coliform and Eschericia coli (E.coli) concentrations within the Yellow Creek Ravine watershed corridor and Lake Ontario shoreline over a decadal (10-year) horizon. Utilizing edge telemetry data collected via the newly integrated Raspberry Pi Zero 2 W platform with triple-isolated Atlas Scientific EZO-circuitry, our predictive model utilizes advanced quantum statistical learning to provide optimal sensor topological placement, minimizing signal reconstruction error and optimizing passive bioremediation feedback loop timing.

1. Basis of 10-Year Decadal CFU Projections

Long-term forecasting of Colony Forming Units (CFU) per 100mL of water samples is vital for municipal recreational safety and ecosystem health diagnostics. Traditional interpolation models fail to represent environmental thresholds under changing climates. Our projection accounts for three highly coupled parameters:

- **Baseline Municipal and Storm Runoff:** Combined sewer overflow (CSO) discharge risks, surface runoff flows, and urbanization factors scale exponentially with local precipitation index *P(t)* and catchment imperviousness factor η.
- **Arrhenius Microbial Heat Kinetics:** Environmental temperature profiles dictate bacterial multiplication kinetics. The model factors in Arrhenius temperature-dependent multiplier $e^{-E_a/RT}$ modeling, anticipating summer heatwaves and heat island shifts.
- **Optimal Spatial Topology Decay:** High-density placement models select coords based on spectral concentration. Real-time feedback control loops deploy active aeration and passive bio-retention filters to decay microbial concentration below risk thresholds.

Unmitigated Baseline Equation:

$$\text{CFU\_baseline}(t) = 480.0 + 35.0 * t + 4.2 * t^{1.35} + N(0, \sigma^2)$$

QML-Guided Restoration Mitigation Equation:

$$\text{CFU\_QML}(t) = 25.0 + (\text{CFU\_initial} - 25.0) * e^{-\lambda_{opt} * t} + N(0, \sigma^2)$$

2. Exact Quantum Machine Learning (QML) Model Architecture

To resolve non-linear spatio-temporal correlations without overfitting, traditional Radial Basis Function (RBF) kernels are replaced with a hybrid **Quantum Kernel-Enhanced Gaussian Process Regressor (QK-GPR)** solver. The model is defined by three primary steps:

- **IQP Feature Mapping:** Multimodal edge telemetry covariates (Conductivity, pH, ORP, Temperature, IMX462 optical night-vision turbidity indicators) are mapped into high-dimensional Hilbert state space via an Instantaneous Quantum Polynomial-time (IQP) ansatz:

$$U_\Phi(x) = \exp[\, i * \Sigma_j\, x_j\, Z_j + i * \Sigma_{j,k}\, x_j\, x_k\, Z_j\, Z_k \,]$$

• **Variational Kernel Overlap Matrix:** A variational ansatz circuit $V(\theta)$ composed of Multi-layer Chebyshev rotation gates extracts cross-correlation patterns. The state overlap is computed directly on physical qubits:

$$K(x_i, x_j) = |\langle 0^{\otimes n}| U_\Phi^\dagger(x_i) V(\theta)^\dagger V(\theta) U_\Phi(x_j) |0^{\otimes n}\rangle|^2$$

• **Log-Likelihood Optimization:** Classical optimization loops tune variational parameters θ to maximize marginal log-marginal-likelihood, allowing the regressor to accurately predict decade-long CFU trends with a reconstruction target error of < 8.0%.